

Antreas Antoniou

AI Research Scientist & Engineer

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Research interests

Learning systems that adapt with limited data, compute and memory. My research connects meta-learning, generative modelling, multimodal representations and evaluation with the engineering needed to train and deploy models. Current directions include persistent memory, richer temporal learning signals and world models that support internal simulation and action.

Education

2017–2021 · PhD in Machine Learning, University of Edinburgh.

2016–2017 · MScR in Data Science, Distinction, University of Edinburgh.

2014–2015 · MSc in Data Science, Distinction, Lancaster University.

2011–2014 · BEng (Hons) in Computer Systems Engineering, Lancaster University.

Research & professional appointments

Chief Science Officer / Head of Research — Axiotic AI

November 2025–present

Conceive and develop frontier-AI research programmes, architectures, training systems and evaluation methods. Hands-on work spans model training, experimental analysis, GCP infrastructure, a self-built home GPU cluster and agent-driven research campaigns, alongside scientific strategy and research direction.

Principal Research Scientist & Head of ML Research — Pieces for Developers

March 2025–March 2026

Conceived, designed, implemented, trained and optimised small language and embedding models through to on-device delivery. Research also covered retrieval and memory, evolutionary deep-research systems and large-context small models. Confidential work described at technique level.

Senior Machine Learning Research Scientist / Engineer — Malted AI

May 2024–February 2025

Designed and built model-training and experimentation systems for small-data LLM adaptation, synthetic-data generation, teacher committees, evolutionary prompt optimisation and distillation. Developed the code and learning recipes supporting product development.

Research Associate — University of Edinburgh

2021–2024 · Data-efficient, transferable and robust generalisation learning

Research on multimodal data, representation learning and evaluation; research software and infrastructure development; MSc research design and supervision; technical support for collaborating researchers.

Research Intern — Google

2020–2021 · Few-shot learning

Investigated the transferability of few-shot learning systems under extreme domain shift.

Selected research & engineering contributions

Model training and research infrastructure — Axiotic

- **DAEDALUS multimodal world-model series — 0.5B, 1B, 4B, 8B, 16B, 32B and 72B parameters:** persistent memory, latent reasoning, internal simulation and compute-aware learning. Designed its self-improvement flywheel around user interaction and autonomous agent testing across versions, using Byzantine validation methods researched at Axiotic.
- **Largest model in the series (2026):** trained the 72B model from scratch on approximately 1T tokens using 64 H100/B100 GPUs on GCP (PyTorch, Accelerate, FSDP); owned run design, training performance and execution.
- **CHRONOS:** designed experiments on learning signals beyond next-token prediction; personally trained a model ladder to 536M parameters and 5.5B tokens on one RTX 4090.
- **Adamantium:** designed and executed architecture/training campaigns comparing BPE transformers, byte-convolutional and hybrid embedding models. Recorded arms span **5.8–9.4M parameters and approximately 10B teacher-token exposure per run across repeated corpus passes**. Developed fast-screening and full-benchmark evaluation workflows; investigated benchmark comparability, data truncation and provenance failures.
- **GCP research infrastructure — Olympus:** designed and built a GKE GPU platform with Kueue workload admission, Spot GPU workers, Workload Identity and immutable, generation-pinned GCS experiment storage. Integrated agents and campaign orchestration with training, evaluation and run-recovery tooling.
- **Home GPU cluster:** designed and built the physical hardware from scratch, then configured a two-node, five-GPU k3s cluster (4× RTX 4090, 1× RTX 5090) with Kueue scheduling for model training and evaluation.

On-device models and memory systems — Pieces

Designed, built and trained 20–80M-parameter language and embedding models through to on-device delivery, including failure diagnosis and optimisation. Research included retrieval, evolutionary report generation and small-model context expansion. Internal metrics are not disclosed.

Adaptation, synthetic data and distillation — Malted

- Built configurable frameworks for Llama 3.1, Phi 3.5 mini and Qwen2 adaptation using LoRA, QLoRA and full fine-tuning, including experiments with 32–2,048 examples.
- Designed teacher-committee synthetic-data generation and selection methods, and ensemble decision-making experiments using averaging, threshold voting and learned integration.
- Developed distillation experiments from 8B models into sub-600M models, with code supporting architecture, pretraining-task, dataset and optimisation choices.
- Conceived and implemented an evolutionary prompt-optimisation system for an 8B model in one day. Detailed recipes and results remain under NDA.

Meta-learning, generative modelling and evaluation — Edinburgh

MAML++ and learned self-critique: developed learning methods, training code and experiments for adaptation from limited labelled data; investigated learning with unlabelled examples through self-critique and augmentation-based task construction.

Data Augmentation GANs: developed generative models that learn augmentation distributions transferable to new classes. Related work includes relational reasoning and representation learning.

GATE/EEVEE and TALI: developed multi-domain, multi-task, multimodal benchmarking/evaluation tooling and a temporal audio–language–image/video dataset, connecting data construction with evaluation beyond a single benchmark.

Research conception, supervision & technical development

Machine Learning Practical and MSc research — University of Edinburgh

Conceived and pitched much of a multi-year portfolio of research projects to student groups and MSc researchers. Contributions extended through architecture and objective design, experimental planning, implementation support, supplied neural modules, training diagnosis and manuscript development. Worked with approximately 10–15 MLP teams in a typical year; individual projects combined my research direction with students' implementation and investigation.

World models, reinforcement learning and transfer

- **Small World Models — David Ivorra-Piqueres and Martin-Philipp Irsch (2019):** research guidance and review on model-based reinforcement learning in CarRacing under limited compute.
- **DARLA/CoinRun (2021):** project work examining disentangled representations, beta-VAE-based learning and reinforcement-learning generalisation.
- Broader project portfolio included deep reinforcement learning, reward design and transfer to unseen settings. These projects developed expertise in evaluating learned representations and policies beyond their training distribution.

Multimodal generation and relational reasoning

- **CLIP-guided text-to-face generation:** architecture guidance on combining AttnGAN with CLIP sentence embeddings and image–text similarity objectives; comparisons of encoder and loss variants and evaluation using image quality and semantic consistency.
- **Relational image-to-image translation:** supplied neural modules and developed architecture/experiment guidance for CycleGAN-style translation with relational discriminators, including generator/discriminator ablations and evaluation design.
- **Low-data domain translation — Jingji Yu:** MSc project on reusing pretrained representations and generators, including CLIP and StyleGAN2, to study data- and compute-efficient translation.
- **DamageGAN:** guided synthetic-augmentation evaluation using image-to-image translation, downstream classifiers, repeated trials and controls separating augmentation effects from transfer learning.

Augmentation, curricula and robust learning

- **EvolutionAugment — Srihari Humbarwadi:** research design around sampling augmentation policies, evaluating their difficulty and training on selected hard examples; comparisons of generalisation and computational cost.
- **Adversarial augmentation — Arpit Rai:** research on optimising transformations and focal perturbations for robust learning. Contributed historical augmentation code covering affine transformations, blur and noise as a technical project resource.
- **Dynamic curricula for class imbalance — Sandhya Kulkarni:** design and supervision of loss-driven sampling and curriculum updates, with comparisons against random and class-balanced baselines.
- **Medical multitask learning:** authored technical review and manuscript edits examining auxiliary localisation as a signal for classification/generalisation; contributed to collaborative medical-imaging research.

Research development beyond coursework

Invited selected teams to continue promising projects towards research papers, providing follow-up experiment plans, manuscript feedback and publication-oriented discussion. Course-project development, manuscripts and published papers are distinguished in the publication record; not all follow-up projects resulted in publications.

Teaching, technical leadership & service

Teaching and mentorship

- **Machine Learning Practical, Edinburgh (from 2017):** lead TA, associate lecturer, group tutor, demonstrator and online instructor across different periods; supported approximately 1,500 students. Developed teaching materials, supervised research projects and managed course cloud infrastructure. Lead-TA appointment: 2017–2020.
- **MSc research (2021–2023):** main supervisory contact for four students, with regular research discussion, technical troubleshooting and feedback on experimental design and dissertations.
- **Model compression (2025–2026):** delivered the LLM-101 model-compression lecture for Edinburgh's Machine Learning Systems course, covering transformer foundations, fine-tuning, pruning, quantisation, distillation and deployment.
- **GenAI as Creative Companion (2025–2026):** created an interactive lecture and reusable teaching package on human-directed AI co-creation and agentic workflows.
- **Invited talks (2025):** GenAI Superpowers at Teach-A-Thon; LLMs for Teaching; What I Learned after 3000 Hours Using ChatGPT.
- **Accelerated deep-learning course (2021–2022):** designed and delivered six weeks of individual instruction; also ran research-group onboarding and technical teaching sessions.

Infrastructure and research community

- Built a research-group Kubernetes cluster, authored [kubejobs](#), and developed the [Minimal-ML-Template](#) research framework. Created documentation and tutorials for GPU-based research workflows.
- Early user and community organiser for Edinburgh's **EIDF A100 GPU cluster**: founded its Slack community, wrote documentation and coordinated user/developer discussions.
- Rewrote medical-imaging research code for the HDR UK project and provided continuing engineering support; supported POEM research through experiment orchestration and technical discussion.
- Procured and set up a **£50,000 deep-learning research server**, including hardware comparison and vendor negotiation. Secured **US\$20,000 in GCP research credits (2021–2023)** and a **Google TRC TPU grant (2021)**; managed MLP course cloud resources.
- **LeRobot Edinburgh (2025):** conceived and initiated the local global-hackathon event, assembled the organising team, supplied robotics/GPU hardware and mentored teams during the competition; team achieved seventh place internationally.
- **Workshop on Efficient LLMs (2024):** key organiser; GAIL funding awarded.

Recognition

2026: selected expert, European Expert Forum on Frontier AI, European AI Office. **2023:** Sustained Excellence Contribution Reward, Edinburgh. **2021:** School Staff Award for MLP contributions. **2019:** teaching-award nominations; finalist, Best UK PhD Tutor; top-three finalist, UK Open Source Awards (MAML++). **2018:** Best Student Who Tutors nomination. **2015:** IBM Prize for Best Data Science Dissertation. **2014:** MSc scholarship; second place, Lancaster CS Hackathon.

Earlier appointments and technical foundations

Amazon (2016): Speech-Scientist Intern, Amazon Echo. **Lancaster (2015):** Research Associate, Deep Online Cognition/Dana; Digital Innovation TA. **Lancaster (2014):** Embedded Systems Research Intern, hardware and control programming for Blackpool Illuminations. **Lancaster (2013):** Software Developer Intern, interactive Android application.

Technical: Python, PyTorch, TensorFlow, Keras; distributed training, GCP, Docker, Kubernetes; Java, C/C++, LaTeX; electronics, embedded systems and signal processing. **Languages:** Greek (native), English (proficient), Japanese (basic).

Publications — conference and journal papers

1. Linus Ericsson, Miguel Espinosa, Chenhongyi Yang, **Antreas Antoniou**, Amos Storkey, Shay B. Cohen, Steven McDonagh and Elliot J. Crowley. **einspace: Searching for Neural Architectures from Fundamental Operations**. NeurIPS, 2024.
2. Alessandro Fontanella, Wenwen Li, Grant Mair, **Antreas Antoniou**, Eleanor Platt, Paul Armitage, Emanuele Trucco, Joanna M. Wardlaw and Amos Storkey. **Development of a deep learning method to identify acute ischaemic stroke lesions on brain CT**. Stroke and Vascular Neurology, 2024.
3. Alessandro Fontanella, **Antreas Antoniou**, Wenwen Li, Joanna Wardlaw, Grant Mair, Emanuele Trucco and Amos Storkey. **ACAT: Adversarial Counterfactual Attention for Classification and Detection in Medical Imaging**. ICML, 2023.
4. Adam Jelley, Amos Storkey, **Antreas Antoniou** and Sam Devlin. **Contrastive Meta-Learning for Partially Observable Few-Shot Learning**. ICLR, 2023.
5. Timothy Hospedales, **Antreas Antoniou**, Paul Micaelli and Amos Storkey. **Meta-learning in neural networks: A survey**. IEEE Transactions on Pattern Analysis and Machine Intelligence, 44(9), 5149–5169, 2022. [Publisher version](#).
6. **Antreas Antoniou** and Amos J. Storkey. **Learning to Learn by Self-Critique**. NeurIPS, 2019, pp. 9936–9946.
7. **Antreas Antoniou**, Harrison Edwards and Amos Storkey. **How to Train Your MAML**. ICLR, 2019. Preprint, 2018.
8. **Antreas Antoniou**, Amos Storkey and Harrison Edwards. **Data Augmentation Generative Adversarial Networks**. ICANN, 2018. Preprint, 2017.
9. **Antreas Antoniou** and Plamen Angelov. **A General Purpose Intelligent Surveillance System for Mobile Devices Using Deep Learning**. IJCNN, 2016, pp. 2879–2886.

Workshop papers

10. Fady Rezk, **Antreas Antoniou**, Henry Gouk and Timothy Hospedales. **Liouna: Biologically Plausible Learning for Efficient Pre-Training of Transferrable Deep Models**. WANT workshop at ICML, 2024.
11. Fady Rezk, **Antreas Antoniou**, Henry Gouk and Timothy Hospedales. **Is Scaling Learned Optimizers Worth It? Evaluating the Value of VeLO’s 4000 TPU Months**. I Can’t Believe It’s Not Better workshop, NeurIPS 2023. PMLR 239:65–83.
12. **Antreas Antoniou**, Massimiliano Patacchiola, Mateusz Ochal and Amos Storkey. **Defining Benchmarks for Continual Few-Shot Learning**. Meta-Learning Workshop, NeurIPS, 2020.

Preprints, research resources & thesis

13. **Antreas Antoniou**, Eleni Triantafillou, Hugo Larochelle, Sebastien Montella, Fady Rezk, Kiyoon Kim, Linus Ericsson, Pavlos Vougiouklis, Justin Engelmänn, Elliot J. Crowley, Srihari Humbarwadi, Yi Liu, Guang Yang, Jeff Z. Pan and Amos Storkey. **EEVEE and GATE: Finding the Right Benchmarks and How to Run Them Seamlessly**. 2024. [OpenReview](#).
14. **Antreas Antoniou**, Eleni Triantafillou, Hugo Larochelle, Sebastien Montella, Fady Rezk and Amos Storkey. **TALI: A Dataset of Temporal Audio Language and Images**. Dataset, 2024. [Hugging Face](#).
15. Kiyoon Kim, Shreyank N. Gowda, Panagiotis Eustratiadis, **Antreas Antoniou** and Robert B. Fisher. **Adversarial Augmentation Training Makes Action Recognition Models More Robust to Realistic Video Distribution Shifts**. 2024. [arXiv:2401.11406](#).
16. Alessandro Fontanella, Wenwen Li, Grant Mair, **Antreas Antoniou**, Eleanor Platt, Chloe Martin, Paul Armitage, Emanuele Trucco, Joanna Wardlaw and Amos Storkey. **Challenges of Building Medical Image Datasets for Development of Deep Learning Software in Stroke**. 2023. [arXiv:2309.15081](#).
17. **Antreas Antoniou**. **Meta Learning for Supervised and Unsupervised Few-Shot Learning**. PhD thesis, University of Edinburgh, 2021.
18. **Antreas Antoniou** and Amos Storkey. **Assume, Augment and Learn: Unsupervised Few-Shot Meta-Learning via Random Labels and Data Augmentation**. 2019. [arXiv:1902.09884](#).
19. **Antreas Antoniou**, Agnieszka Słowik, Elliot J. Crowley and Amos Storkey. **Dilated Densenets for Relational Reasoning**. 2018. [arXiv:1811.00410](#).
20. Luke N. Darlow, Elliot J. Crowley, **Antreas Antoniou** and Amos J. Storkey. **CINIC-10 Is Not ImageNet or CIFAR-10**. 2018. [arXiv:1810.03505](#).

Other scholarly work

21. **Antreas Antoniou**, Nick Pawlowski, Jack Turner, James Owers, Joseph Mellor and Elliot J. Crowley. **Meta-meta-learning for Neural Architecture Search through arXiv Descent**. SIGBOVIK, 2019.

Research software, deployed systems & public communication

- **Experiment Framework**: built portable local/remote experiment tooling for pinned-source execution, checkpoint/resume, versioned datasets, W&B/Hugging Face integration and result reconciliation.
- **Cliphord**: built Swift/macOS semantic search using [Ogma models](#) I trained at Axiotic, deployed through CoreML; encrypted storage, model installation and tokenizer/output parity tests.
- **HephaestusZero**: conceived, designed and built a deployed multi-agent runtime with [information-theoretic Byzantine](#) methods: uncertainty-scaled redundancy, diverse reasoning paths and adversarial verification. Layered memory formation and agentic retrieval combine standing facts/preferences/episodes with exact, tag, BM25 and optional embedding search. Multi-resolution storage, access-sensitive decay and budgeted assembly provide situation-relevant context.
- **Archivum and Argus**: designed and built Git-backed research continuity and context-routing tools, preserving project evidence and provenance across sessions while separating knowledge, commitments and resource control.

How to Train Your MAML, **Minimal-ML-Template** and **kubejobs** provide reusable research and infrastructure tooling. Code and teaching resources: [GitHub](#) and [Teaching](#).

Public writing includes *Beyond the Cloud*, *Too Much of a Good Thing* and *AI Memory Explained*, originally published by Pieces. Research direction and further material: [antreas.io](#) and [DAEDALUS](#).